

ActInf GuestStream 045.1 ~ Ramstead & Albarracin: "The inner screen model of consciousness"

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hello and welcome it's June 19 2023 we're going to have a presentation and a discussion with Maxwell ramstead and Mala aburassen here so thank you Maxwell and Mal for joining looking forward to your presentation setting some of the context and then thank you to all of our guests you'll have an opportunity to introduce yourself and give some background when you first uh give Reflections so onto the slides Maxwell excellent well maybe we now and I can just briefly reintroduce ourselves for uh those of us watching at home um so hey folks uh my name is Maxwell ramstead I am the uh director of research at versus AI um I am a scholar in the tradition of active inference and an expert on the free energy principle um yeah and now would you please sure so my name is I work at versus with Maxwell and director director of product for research and development but I'm also doing a PhD in cognitive Computing I worked in social sciences and in artificial intelligence in the artificial intelligence field and I've specifically tried to apply active inference to more socially grounded questions and sometimes more abstract and representational questions diving all the way down to the philosophy that can be attached to it thank you Val um so yeah I'm very pleased to present this work uh today uh Mal and I will be discussing our new pre-print um called the inner screen model of Consciousness and it is about it is basically an application of the free energy principle to the uh problem uh the conundrum of Consciousness um so uh just to give a little bit of background and now I'll just be walking through the slides if at any point you want to uh jump in and you know add complete correct uh please feel free to do so um so the uh this is work uh that started in late last year um and continued throughout the year it's preoccupied us quite a bit actually uh over the last while um now and I are the lead authors on this uh what has now become a series of papers uh with uh one more pre-print coming out hopefully before the end of the months and another in the works hope Pro hopefully before the end of the year uh with the same Core group of authors uh so obviously uh Mao uh as well as Adam saffron Alex Kiefer Brandon Klein Chris Fields uh and Carl friston um and uh the follow-up papers uh

include a bunch of other authors uh so we've been uh working with some uh pretty cool folks on this uh folks listed here uh and others um yeah and so the overall Lane across the uh two or three papers uh is to present something like a minimal unifying model uh of Consciousness premise on the free energy principle in the paper that we're discussing today uh we are presenting as I was uh intimating a moment ago a model of Consciousness that follows directly from applying the free energy principle to well understood human neurophysiology um the take-home message of today's talk is that uh well let me take one step back there has been a lot of discussion in the literature over the last decade uh about whether the free energy principle has anything uh unique to teach us about Consciousness and there have been responses Pro and Contra um and I think that the question up until recently was just unsettled uh we would like to suggest a positive response to that question um and so as such one of the take-home messages is that actually yes a model of Consciousness can be directly derived from the free energy principle uh applied to explain neuroanatomy and um I guess the the kind of technical message the technical part of the takeaway is that Consciousness has or um is isomorphic to a math a nested holographic mathematical structure um so uh this sounds like a mouthful might sound a little crazy um we are going to unpack this uh presently um I'll take some some now did you want to add anything to that no all good excellent um so uh the original paper uh that we wrote turned into like a 24 000 word monstrosity of a paper uh and it was uh elephant elephantine both in the sense that it was very large um and also in the sense that of the uh Sufi parable of the Blind Men and the Elephant what we're trying to do is in some sense um identify a parsimonious mathematical structure that underwrites the different accounts of Consciousness that have been developed based on the free energy principle I'll spare you the original paper structure um I said a moment ago that we uh we're developing a minimal unifying model this is a notion that was proposed by our friend and colleague Vania Visa uh uh I believe a few years ago now in 2020 um and Vanya at the time was noticing um that well the field of Consciousness studies is replete with a bunch of different models and theories um and there seems to be no end in sight to the multiplication of these theories and uh you know the idea is maybe what the field needs is not one additional you know n plus one Theory Of Consciousness but rather a minimal unifying model uh by which Vanya means a model that specifies only necessary properties of Consciousness so this doesn't come with like a strong sufficiency claim really like what are the kind of rudimentary building blocks that are necessary for Consciousness in general um this model would have determinable descriptions that can be made more specific so it's kind of a base model that can be expanded to cover

different varieties of conscious experience and finally that you know it's minimal and unifying in the sense that it integrates a current approaches to Consciousness in no small part by highlighting their common assumptions um so you know kind of pointing to a least common denominator of all existing accounts and so what we have tried to do in this paper is to uh well in this series of papers is to do uh engage in this analysis and this uh mummy analysis if you will for theories of Consciousness uh premised on the free energy principle directly so it's worth quickly discussing the free energy principle uh I'm almost embarrassed to do this at the Active inference Institute uh but it's always worth going through this very quickly um so there's a standard distinction in physics that one can leverage between Dynamics mechanics and principles the free energy Principle as it says on the tin is a principle um so principles are kind of the foundation of a hierarchy of theory building in science at the as it were top of the hierarchy uh our formal descriptions of behavior what are known as Dynamics so if you're applying dynamical systems theory to understand uh you know the time evolution of some system you're in the realm of Dynamics uh arguably folks like Kepler and Galileo were also in the realm of Dynamics so I see some empirical phenomenon and I produce a formal description of that behavior mechanics comes into play when we move from Kepler and Galileo to Newton when we move from merely describing some Behavior to providing equations of motion that allow us to explain why that behavior has the particular shape that it does um so uh you know classical Mechanics for example gives us an account of gravitational force and explains why orbits are shaped the way they do so if if Galileo and Kepler just noticed the shape of orbits and provided a formal description of them Newton explains where that shape comes from in some sense and principles in turn explain where mechanics come from so classical mechanics come from the principle of least action or stationary action which uh very roughly speaking is the principle According to which uh and no more or no less energy is used than necessary to perform some physical movements that kinetic and potential energies balance out such that the true Paths of the system through its state space are those for which that balance equals zero the principle of least action uh explains where uh classical mechanics comes from in some sense and similarly uh the free energy principle is at the basis of another kind of mechanics uh that is becoming known as Bayesian mechanics you can think of the 300g principle as having the same relation to Bayesian mechanics as the principle of least action has the classical mechanics and Bayesian mechanics is the physics of probabilistic beliefs so um it's a physics that connects the uh properties of the system that you're considering the physical properties of the system so for in particular the information entropy

of uh states of the system to the uh thermodynamics sorry to the in it connects the thermodynamic entropy of the states of a system to the information entropy of the beliefs that a system has about the the uh systems to which it is coupled here A probabilistic belief uh means a probability density uh uh in a very general sense we're not assuming some kind of contentful notion of belief like you might find in uh philosophy really this is about kind of a tracking relation but we'll get that to that in uh actually just a few slides there are two main formulations of the free energy principle in the literature the classical and the quantum both rest on this apparatus of the Markov blanket uh the Markov blanket definitionally is uh the degrees of freedom that separate but couple uh two systems or rather two things within a larger system it's effectively a statistical boundary uh that allows us to say that well given this boundary the inside is independent of the outside um the way that we uh Express the dependencies so taking one step back quickly this is a dynamic systems approach to uh to cognition to a self-organizing systems and to physical systems more generally uh it is dynamical systems approach in the sense that we're you know using the tools of dynamical systems theory to examine the time evolution of a system it's trajectory in its state space and in this context a generative model is used to encode the relations of dependents that obtain in the time evolution of the system so this is important and I've been uh you know uh the kind of making this point uh like uh maybe Ad nauseam for the last I guess half decade but the generative model is not a model that you have in your head it is a uh it is a statistical model that represents the conditional dependence structure of the entire system that you're considering so this is an an environment agent system uh in the broadest sense um and uh the free energy principle says that if a specific sparseness structure obtains uh so if things are disconnected in a special way uh then uh it'll look as if uh the subsets that are disconnected are tracking each other um so uh more technically uh we partition our system into particles uh particles are basically internal States shrouded behind their Markov blanket where the Markov blanket is composed of sensory States and active States sensory states are affect affect internal States but are not affected by the internal States and active States affect external state but are not affected by external States uh this uh kind of this separation of causal dependence uh and this structure uh intervening between the two bulk uh subsets of the system is really critical um the free energy principle says if if my generative model which encodes all of the dependencies of the system if my generative model contains a Markov blanket in the sense that we just defined then we uh are basically licensed in our interpretation of internal States as basically tracking external states where tracking has a specific mathematical definition in this

context tracking means to encode the parameters of some probabilistic belief about external States so to the extent that we want to call this representation we should note that this is not a representation in the typical teliosesemantic sense of having some internal States track some external features um it what we're what we're saying is that basically if I have a Markov blanket then my internal states are basically shaping a belief about the external world and I think that's really critical to understand um and this is where the inference and the modeling that comes from this kind of tracking relationship formally speaking is approximately inference so basically in a nutshell the free energy principle says in our physical universe as we understand it with the the usual mechanics running in the background right classical relativistic Quantum and statistical uh if boundaries exist in the physical system then the bulk across the boundary is going to be tracking whatever is beyond the boundary so in a nutshell this is what the free energy principle says now I noticed you put some stuff in the chat would you mind using your voice to complement that yeah I was just saying that this part that you were just explaining often seems kind of as etheric to people um but it's critical to understand the sparseness of the coupling because then you kind of start understanding where perspective comes from and if you have perspective you also have uh the the capacity to um to assess causality um starting from a point and extending outwards to something else and this is true from any so that's that's what I wanted to add thank you Mama I think that's useful addition um yeah so this is the classical formulation of the free energy principle um there is a new hit funky uh uh formulation that has become available over the last half decade due to uh Chris Fields uh Jim glazebrook and their colleagues these so-called Quantum information theoretic formulation of the fep fear not this doesn't appeal directly to Quantum Mechanics rather it appeals to uh the tools that have been used to extend information theory in the context of quantum mechanics which allow you to calculate like all sorts of funky wave equations and propagation operators and all that we don't have to get into the details basically it's just a way to reframe what I just said so what I just said is if there's a boundary in my physical system then the things across the boundary will look as if they're tracking each other another way of saying that is um well this boundary is a kind of screen uh green uh takes on a particular sense in Quantum Quantum information uh Theory uh we don't necessarily have to get into the details of it but basically um you can think of these degrees of freedom that couple the two systems as a kind of probabilistic surface and you can think of the bulk across either side of the boundary as in alternation reading and writing information onto the blanket so uh any Markov blanket whatsoever can be construed as a classical information Channel or a screen

and in effect uh all of the classical information that you need to describe the way that things across the boundary so the so-called bulk of the system couple to uh one another is contained on the boundary it encodes the classical information that's necessary to describe the couplings uh within the system um so uh we this is the this is holography not in the sense of um this is the Hologram not in the sense of holography although we might discuss senses in which these things are intertwined um we're we're using the word hologram in the sense of the holographic principle uh from physics the holographic principle is a principle that was originally uh discovered in the context of black hole thermodynamics and what it basically tells us is necessarily from the point of view of an external Observer some bulk in in physical space can only contain as much information as can fit onto its surface uh so this Echoes what we were just discussing about Markov blankets the physical reason is that if a bulb collapses into a black hole um and the bulk contained any more information than could be fit onto its boundary uh then we would basically be losing information and that would violate uh the conservation of classical information also known as the principle of unitarity in quantum mechanics which basically says that information is never lost um so uh we're using the word hologram in that sense this is different from the sense of hologram uh I mean everyone has seen like the Hologram of Tupac or does that make me look seem old at the Super Bowl a few years ago or maybe Pokemon cards maybe that also makes me seem old now I don't know what the kids are doing but you've seen like red baseball cards and stuff like that with different superimposed like picture layers that become visible as you move that's not the sense of hologram that's at stake although maybe so we'll get to why but let's we'll focus on the Hologram theoretic formulation so to summarize in the Hologram theoretic hologram holographic principle version of the free energy principle the quantum information theoretic version of the principle all of the classical information that I need as an external Observer to characterize the coupling between two subsystems of a larger system exist on the boundary um and that's a pretty remarkable physical fact and a few years ago uh Chris Fields Jim glazebrook and Mike Levin proposed that maybe uh this kind of uh information concentrating bottleneck architecture uh also has application to uh Consciousness in the form of an internal screen so the idea was that you know the the information that I bring to bear to parse my sensory stream and make sense of what is constantly bombarding me it has to live somewhere and so uh the idea is well it lives in this inner screen so you basically have an external Markov blanket which is a Markov blanket just in the usual sense and then um the idea is well within the set of internal States um that uh are kind of partitioned off from the organisms due to the organisms Markov blanket

then you would have an internal Markov blanket structure within those internal States um this is prediction six from the great paper minimal physicalism internal awareness requires internal boundaries and there's this interesting connection to integrated information Theory I'm I I am agnostic with respect to IIT so we're not going to go there today but for those of you who are interested this kind of inner loop mechanism seems to generate positive integrated information so there's an interesting connection there um before I move on to neurophysiology I'll hand things over to Mao if you you wanted to add anything yeah actually it was literally about to type something um that this is actually also connected to some Neo materialist formulations of um folding onto onto itself so maybe this is something worth digging into a little later in the talk awesome uh then we shall and I I believe that Ali is also a co-author uh on the paper and we have him so uh Congratulations by the way on the pre-print I I saw it released and Mao has told me about it I'm really happy to see uh such serious uh philosophical work being done with the free energy Principle as the basis I think you should be very happy with the end result and proud of but uh now uh let me get back to uh my monologue um so our uh rather uh our now and my uh diet it's not really a dialogue it's sort of like a I don't know how to call it anyway um I'm thinking in French I have the words in French I promise um so um now that we've you know reviewed in a very much too short way to brutally fast fashion uh the free energy principle and its core formulations uh we're now going to derive a model of Consciousness by applying it to the known neurophysiology of the brain are you ready you buckled up okay good um so uh we have just said that the fep can be used to model things that have boundaries I submit to you that the brain is a thing with boundaries and importantly a lot many several internal boundaries um so in particular we know that the brain has a very sparse structure um so you know if you think of uh just the raw numbers right there are roughly on the order of 150 160 billion neurons in the brain um but each neuron only makes about uh you know something like a five I want to say five figures because I've spent uh so uh 10 10 to the uh five uh connections right so like roughly like I think that my you know my neurophysiology is a bit far behind me but it's something on the order of like a few thousand to maybe tens of thousands of connections per neuron so obviously if you just think about it most neurons are not connected to each other not directly and furthermore the brain has a hierarchical structure and when you think about it what is a hierarchy a hierarchy is a sparse connectivity structure right so if you think about a hierarchy of connectivity what having a hierarchical structure means at the end of the day this isn't some statement about power dynamics where things at the top of the hierarchy are more important than things at the bottom your cortical neurons

don't drain more juice uh than you know other uh parts of the brain the neurons in those parts uh no what we're talking about is a regular sparse pattern of connectivity where I only make connections to the layers directly below me and to the layers directly above me what I want to suggest now is that you can understand the sparseness as a set of nested Markov blankets and that's going to do a lot of heavy lifting for us um so the discussions here are based on two papers uh one by uh fields and colleagues called neurons as hierarchies of quantum reference frames it's really cool I recommend it uh and also a paper by inesh ipolito myself and our colleagues on Markov blankets in the brain so this is going to be a much too short summary of those discussions but basically you can start with individual neurons and then um you know talk about uh their Markov blankets um to bring it back to something that I was saying earlier but that I didn't emphasize while we were going through the slides the Markov blanket identifies dynamical dependencies not necessarily physical boundaries so it's a statistical boundary it's not necessarily like the boundaries of a wall like a cell in the case of formalizing individual neurons with Markov blankets we would associate the postsynaptic conductance with the internal States the post-synaptic voltage with the active States the pre-synaptic conductance to external States and the pre-synaptic voltage to sensory States you have some equations of motion that tell you uh how each of these quantities change and what they depend on and you'll notice that this has the uh the dependency structure of a Markov blanket where the internal states only depend on uh sensory active and internal States um the active states only depend on internal States and so on so it respects the dependency structures uh that we were discussing a bit earlier where again you know the sensory states can only depend on other blanket States and external States and so on so here we have a nice Dynamic Markov blanket that we carve around our neurons and then there are two directions that we can go we can Ascend the scales uh towards uh you know canonical micro circuits and then uh you know cortical columns uh individual brain regions brain whole brain networks and so on and at each of these successive scales of self-organization you can similarly identify uh Markov uh if you want to uh uh discuss this in more detail uh you can go see uh the paper by inesh and myself and colleagues um and you can also go from the individual neuron down uh towards um gated channels towards uh synapses uh dendrites and the uh Soma so the body of the cell all of these things have successive Markov blankets and so all throughout the brain's uh organization at the different scales at which it self organizes you have this nice structure of nested Markov blankets of Markov blankets of Markov blankets um so that's one clear immediate sense in which there are these blankets and you know you can start to think of these blankets as

screens uh in the way that we uh just uh we're talking about a slightly more illustrative example of what we're talking about here is that the martial blankets and predictive coding this is a figure pretty typical in the predictive coding literature what it tells you is that the brain has a series of uh levels or layers where superordinate layers are basically passing predictions about sensory input onto subordinate layers and subordinate layers are shuffling prediction error back up to the superordinate layers this is uh probably like you know I'm almost embarrassed again to discuss this at the Active inference Institute I'm sure everyone is quite familiar with these things already the novelty that we are introducing here is to say well each of these layers is connected to each other via a Markov blanket um and so that's I think like a key insight is that each of these layers has internal States and communicates two other layers through what you can formalize as a Markov blanket so you can understand at any given scale of brain activity you can understand the different levels of message passing as successively nested screens the cool thing is that what you know what each of these screens is doing in some sense is coarse graining the uh stuff in the screen below so in some sense each screen is kind of implicitly contained in the following one and what you end up having is a kind of nested structure of screens of screens of screens of screens which is uh I mean pretty cool we recover as you'll note in this figure the um the inner screen hypothesis each of these uh you know this is probably a better way to visualize it you have to at the left the uh out uh the kind of external Markov blanket of the organism it is uh segregating or carving out a set of internal States and what we're saying is these internal states have a layered level level structure a sparse structure that we can formalize using nested Markov blankets um and like I was saying each screen successively coarse grains uh the next screen and this gives some flesh to this notion of contextual Computing this idea that really what the brain is is an organ of context where each layer is providing context to the layer below and where each layer is attuning to increasingly higher order uh slower and more physically widespread regularities so uh one thing that we're proposing here uh which is a novel hypothesis that we intend to test empirically over the next few years is that this architecture has two special layers in some sense there are two uh you can think about it as right only layers so if you're thinking about the stack um the absolute well the the per the sensory periphery or the sensory bottom it only sense in one sense can only write in One Direction it can only write into the system um and the topmost level um can only write downwards and this is a key architectural feature one thing to point out is that in the paper we Associated uh the topmost level to structures like the hippocampus we don't need to suggest that the hippocampus is uniquely the top uh of this

hierarchy rather there is probably a whole bulk of structures that uh basically do do not cannot any longer be coarse grained so you kind of have like an executive level pop and what's particular about this top is that it can only write down into the stack so it's its function really is to be about and monitor the other layers of the stack implementing a form of covert action and which trickles down uh throughout the stack so um now I saw you write some stuff uh would you like to that yeah so I think one of the things that's important to highlight is why some things can only write up and some things can only like write down um and what this means for irreducible the irreducible market linkage which is the part where if uh where our experience happens uh because there's a counter that and and some future questions that are going to happen that relates to Super structures and then why don't we experience the consciousness of a superstructure so I think it might be important to dig into that a little bit absolutely um thank you uh so yeah uh to summarize then uh there are basically two right only layers uh as you ascend the layers which are successively coarse graining each other uh there is an irreducible screen that can't further be coarse grained uh we believe that this corresponds to something like a naturalized homunculus and we're calling this perspective it with an intentional provocation we're calling this perspective neocartesian um in the sense that uh yeah we we may have in some sense uh identified the processes that make it seem as if there was something like a homunculus obviously uh we're not or maybe not obviously it's worth saying we are not um falling into the homunculus fallacy because we're not claiming that the topmost layers perceive themselves so the buck stops with the top layers uh and the top layers can only perceive themselves vicariously by acting through other layers um and you know we we associate this action of layers of internal Markov blankets on each other with attention in particular and other neural modulatory effects um so I mean the overall picture is one where Consciousness is intrinsically related to overt and covert actions uh to the overt uh acting and sensing associated with the external Marco blanket and with the covert actions that are being deployed within the stack and in particular that are issued by the very top uh set of levels as I said this is not necessarily a strict hierarchy it could be a hierarchy with several different tops but this integrated top structure we believe uh could be associated with a naturalized homunculus I'm going to skip through uh the rest and thank you for your attention uh unless now one wants to add something just one thing is I think to put it in in simpler terms as well the two layers that can only write can't store memories because that's right memories is not um it's not just an encoding it's a process of um of retrieval and it requires interactions between different levels which is why the lowest levels are only interacting with things which are

external and therefore can't necessarily store and we can't like make direct interactions with anything else within the The Entity itself and the highest level as well so there's no memories being encoded at these two levels which is why they're right only and not um yeah I think that that helps a little bit thank you yeah I would agree um so uh so thank you all for your attention thank you in particular Dan for the invitation it's always great to be at uh the idol a double I and uh yeah we uh looking forward to this discussion I think Daniel said he wasn't going to uh to do too much um um animation so uh if people have are you gonna do uh are you gonna do any facilitation at all then yeah okay so uh thanks Maxwell and Mao for the amazing talk um there are so many interesting things to dig on in this paper I don't know even where to start but I do have one question too actually in particular so uh does this in-screen model imply a kind of Proto and psychism or a more restricted bio psychism I mean do you think the type of nested holographic structure uh described by the model only exists in biological systems and regardless of the answer would you think that it generalizes to larger systems comprising these systems like populations and colonies and so on maybe I can take a quick stab at that um so we uh we think that um the free energy principle itself commits you to a kind of um I don't know what the right word is we're kind of we're writing a paper now uh we're struggling with the exact coinage but it's something it's not quite a pan psychism it's it's a pen representationalism or a pen tracking ISM that you'll allow me to absolutely disgusting uh neologism um basically what the fpp implies is that anything that has a boundary is it looks as if it's tracking what's beyond the boundary so at a very basic level the fep kind of licenses this talk about anything so anything whatsoever looks as if it's tracking uh what's across the boundary by virtue of existing um so you know at a very basic level if I see a rock outside I know that you know it's not a thousand degrees Celsius outside right like the the fact that like a rock has a specific structure entails something about its environment uh likewise I can see uh you Maria you are not wearing a winter parka and I can uh you know infer from that that it's probably room temperature around you and not like you're not somewhere in Antarctica or wherever else so uh this kind of basic tracking relationship applies to everything what we've tried to do in this paper is to start you know starts spelling out more explicitly what the difference is between uh you know just rocks and rock stars in some sense like what what is the special kind of structure that you need to add to the Markov blanket and what you'll notice is that you get Consciousness by making this the overall structure more sparse so as as you make it as you you as you make more of the connectivity zero you get all of these weird things that pop up one of which is we think like this this Consciousness structure um malice thought

about this prop maybe more than I have so I'd be curious to get her perspective well so I know that people are a little bit um wary of pan psychism so we usually add the term Proto in front of it uh and so the idea is that we have to deconceptualize what Consciousness means and oftentimes what we mean is something that resembles what humans do uh and we have to be a little bit if we take our formalism Consciousness can become emergent at the scales where humans are where but but it doesn't mean that it doesn't exist at lower scales where uh the inertia or the entropy generated by policies is um is lower so if you think of a rock its policy is pretty much maintain bonds you know I mean like there's nothing else the scale at which some kind of thing exists and changes is much different than you deciding to have coffee but in effect your cells could be something that we could consider exist closer to something that Iraq does and even lower uh just the proteins that make up your system so um if we understand that there is nestedness um and that the the conscious process happens once you have this capacity to to exchange information across layers anything that has this stacking of layers in a nested fashion is in effect proto-conscious but we have to be very very careful about what we load into the the term conscious I think is is mainly the question I'm I'm sharing my screen here this is from a paper on path integrals particular kinds and strange particles uh so it is a typology of different kinds of Markov blankets uh and different kinds of particles um so something like a rock it still has a Markov blanket but it's its Markov blanket is not partitioned into sensory and active states there are no states that have this property that the active states have that they affect the external world that are not affected by it in return a rock just has sensory States if I kick a rock unless it's Radioactive It's not going to act on me it's just going to uh it only has the inference part of active inference it's only kind of changing his State as a response to the forces that are acting on it um active particles are particles that have both sensory and active States and you might think for example of a cell uh a cell it has this kind of structure um particles like ourselves um have additional sparseness in the Markov blanket in particular what we call strange particles um do not the internal states of strange particles do not have direct access to the active states of strange particles and therefore they need to be inferred and this is what gets you planning at the end of the day a strange particle can only exist in uh by kind of inferring itself into existence precisely because there is there are no direct connections between the active and the internal States so as you remove uh connectivity from the overall structure right uh here from here to here you're adding sparseness from here to here you're adding sparseness as as you make the system more and more sparse you get more and more interesting Behavior to emerge uh from the system and you know you can think of the topmost

uh structure that we were describing as an instance of like this broader architecture uh where the topmost layer doesn't have direct access to the active states of the system and therefore it kind of has to go through the entire bulk of the system to infer itself into existence so there's a few hands up uh I don't there's no um it's not on Google where you can see the people who share it oh okay so Ali I think you raised your hand first uh yes thank you again for your crystal clear and really illuminating uh talk uh I've written down three uh questions but maybe I'll begin with uh probably the most basic and uh naive one uh so uh in some of the earlier literature on active inference and the fvp uh there was this emphasis on distinguishing between generative process and generative model but in this paper uh in figure one which you also shown in your slides uh we don't see any generative process in the diagram so where does generative process fit into this diagram or uh because the way I see it uh I mean it's basically somehow included in the generator model or uh I don't know what is the distinction between these two concepts so um you'll have to forgive some of our the earlier literature uh for this there's some terminological inconsistency uh the only two constructs that are at play here are the generative model and the variational density that's encoded by internal States that's it um so the it is not the case that the FTP is about how the generative model models the generative process the a lot of this language comes from uh you know previous Bayesian approaches and in particular the helmholtz machine in the helmholtz machine uh so helmholtz machines are some of the earlier predictive coding machines and then the helmholtz machine basically you have a forward pass that connects the sensory end to this leave the top of the predictive hierarchy and then a backwards pass connecting it uh the uh you know the the uh the like top of the model back to the sensory end and in in a home holds machine you would call the forward pass a recognition model or uh and the backwards passage generative model right the forwards pass is a recognition model because you're inputting some data and then basically the by the time the information trickles up to the top you have recognized what caused the data and the backwards pass is called a generative model for the the reasons that are sometimes invoked in the predictive coding literature uh where you know you're basically uh generating possible sensoria given a certain configuration of States uh so the we don't use the recognition and generative in that sense anymore um and you know I apologize in some previous Publications I I've been a bit sloppy in the way that I use my language and I'd use the term recognition density what's really at stake is uh and I'll share my screen uh this to uh yeah the help make the point uh that's what it says here so you have a you have your generative model right your generative model is a joint probability density so what is a joint probability

density uh well you have a bunch of different uh basically states that your your system uh is defined over um and you have a probability density uh that the shape of which captures the way that uh states can change over time right so like if I change this way then you know I need to change in that way and that's what this shape encodes um and uh what the Feb says is if my generative model has a certain sparseness right if it has Markov blankets then there's this additional probability density that I can Define uh it is parametrized by the internal states of the system and it's called the variational density and this is like my probabilistic best guess about what's causing my external States um at no point do you need to appeal to any other bits of like yeah so generative process when you're actually coding these models up in Matlab or in Python what we call the generative process is just the equations of motion that govern external States so if you actually hand code one of these one day your generative process might include like Newtonian physics and whatever else it's sort of just like what is the process that actually animates the Dynamics of external States um in in some there in some very simple models like in the uh the ones used by Miguel Aguilera and colleagues you're looking at um linear dissipative systems and they don't have any Dynamics right like so basically the the system kind of just it dissipates it's this a linear dissipative system is a damped in Spring physically right so it just goes bring and then it dissipates to the fixed point so similarly uh you may or may not have Dynamics to the external States and that's what gets harnessed in the generative process uh yeah this should be clear yeah sorry man no I mean I think what the question that Ali is asking also has some important correlates that often get asked to us like what's the difference between Consciousness or Dreaming or what's the difference between a dead system that decays or uh being uh psychotic um and I think what Max was saying is our system currently doesn't say anything about your actual connection to the real world other than a thing which is successful at being a thing is probably well uh joined relative to its needs to the world and therefore um your relation to reality is probably um has probably got a better grasp if you are functional for a longer longer amount of time this minus senescence um so yeah does that kind of yes yes thank you thank you both uh it it really makes a lot more sense now thank you so much well uh it's my pleasure and remember like model just means joint probability density here right so uh there's a a criminally underappreciated paper that Carl wrote uh in entropy in 2012 called the free energy principle for biological systems uh which is where a lot of them are called blanket story comes from and in that paper Carl refers to the generative model as just a generative density so it's basically saying there's a there's a underlying probability density it describes the way that the Dynamics are just are being

generated just generally uh and yeah and then you have the uh if if that density has the appropriate structure meaning if it has like zeros in the right places uh then you can Define this this other tracking density uh that approximates the generative um model or density and I also have a follow-up question which I'll ask after that Samuel and thales okay uh thanks uh for the excellent talk uh uh I'm a professor at the State University of Minnesota uh I'm a philosopher cognitive scientists psychologists so I worked mainly on reasoning but by reading your paper what strikes me as most promising is the possibility of solving the the hard problem of Consciousness if if this goes right and progressing means this model but maybe an important part of that depends on exactly what is meant by Holograms and exactly what this brings into a a theory as an explanation right so doing your presentation you were talking something about that there's a difference between holography in the sense of bringing light to something to make a projection uh but you said that might be related as well so I'm quite curious about these how to use this this concept and how to understand it uh in a more Soul psychology sense and also like where it comes from because I'm not a physicist so does it have does it depend on string theory to to to be correct could you dive in a little more on this concept first now I think this is more a you question Theory so basically remember Maxwell really gave a good explanation I kind of thought I had to answer your question but I'm gonna give a little bit so the boundary between any two mutually separation systems A and B is the holographic screen that encodes like the eigenvalues of the interaction that couples A and B so uh this is very similar to how a hologram encodes information about 3D objects onto a 2d surface um so a hologram stores information about the phase and the amplitude of Lights reflected from an object and this allows the Reconstruction of a 3D image from a 2d surface and so here similarly the holographic screen uh encodes information about A System's interaction with its environment which allows for the prediction and control of the systems Behavior so does that yeah but does does it maybe allow for the creation of an image that's that's the sort of question so this is yeah this is where we're giving it some thought actually so the creation of an image is it's a complex idea because what's an image right but effectively there is some work being done into geometry and how that may potentially interact with matter uh in in order to to to translate to transmit information basically uh and so it's possible that the Reconstruction of information is done through Fields being changed such that the modalities you have through your senses such as sight um hearing touch or relative to the way that the information needs to be transmitted in terms of amplitude and um and velocity Etc which in effect gives you something like light um but that research is still relatively early so

yes it is promising it is leading in that direction it's probably one of the most interesting formalism because it touches on the connection between um you know Dynamics matter and how allows for matter to be organized but all those pieces have been properly being put together it is heading there though um I would just add um the so the original monstrosity of a paper had a section on uh yeah explaining where qualia come from based on the statistical patterns of the input but as Mao said like it was very early work and I mean the core model was more or less developed so we decided to like spin that out as a separate thing and then develop you know the the inner con the inner screen model of Consciousness per se um you know to flush out a little bit or to add some flesh to what now it was uh saying which was already pretty meaty as it is um so we we have you heard of these experiments where they they take a little uh they take a little pad with little pins on it and they basically reconstruct well they they they plug a camera on your head and they put the this little pad on your tongue and if they blindfold you and they they basically have you navigate and the um the camera generates a kind of um like grayscale you know version of the image but with the little pins in terms of like how far they push in uh Samuel are you seeing the image that I'm that I'm discussing or so you have a little little pad it has little pins that can push uh at various depths and what what you can do is basically translate the uh image from the camera into a different uh array of like assorted pins and so yeah you put this on your tongue and if you blindfold yourself then use this camera for about 20 to 30 minutes the brain eventually recognizes that oh this is supposed to go to the visual area and then then you start to see through your tongue in some weird way like the the the the body actually recognizes the statistical frequencies of the uh like of the pins actually pushing onto your tongue and and this is experience not um as like a a physical sensation but actually visually like you start to experience an impoverished grayscale visual field just from the sensations that are being fed to your tongue which suggests that there's something deeply in you know deep about the statistical structures of the signals that are being processed and the way that the qualia show up in your experience um so uh yeah uh but this is still like super far off in that I wouldn't I wouldn't have the pretension to claim that we've cracked or call you come from just yet yeah yeah uh I get what you're saying now uh different uh patterns can uh make different quality that you don't need like different Magics to to have different quality but yeah I was wondering how that related to to the Holograms but okay it's something in development as you as you're saying maybe I should uh let someone go and then I'll go back to these Consciousness questions sure so Alice well thank you all it was a great presentation and great test by the way hi Max thank you hi everyone I'm sorry you guys

are facing like a black screen right now my camera is off that's all right okay so um I'm an anthropologist during psychologist so I'm kind of an outsider in here I'm not from the active interest lab too so you know I study it but I'm not a specialist in it so my questions are kind of broad and they're actually really really simple but I think that I'm not allowed to make one question right now and then right so um actually my first question will be uh something that Maria was talking about before me started so um the Articles account Consciousness is an attribute of things that possesses and reducible Markov blanket right with active States exerting for instance over external Dynamics so when applying to the neural uh physiological experience of the human brain talking about neuromodulatory influence over neuronal Dynamics uh and with that in mind I would like to know that should actually speculate a little bit on the implication of this structures issue of Consciousness for how it might emerge during infant or field development that's a great set of questions uh we gave this talk um uh I gave a version of this talk to um Luis Passoa's group at the neuroscience and philosophy salon and Anna Shanika asked a similar question I would um I don't know I mean Mao probably has a better response I would point you in the direction of the work of Anna Shanika um she has some really cool work on co-embodiment and homeostasis and pregnancy and this kind of thing um I uh I confess that I need to think about uh the way of it like because presumably you could just continue extending this outwards right so we've talked about inner screens and there's like an external screen but presumably you could just reapply this whole screen talk in the other direction like on the other side uh yeah exactly the first prior well the the first prior is um one of those papers but she's got like another a set of papers uh yeah it might this might be it right um so one thing to also know could consider is the sparseness aspect of um of our model um as you develop you prune connections and you become more and more sparse which gives you more path approximations to the way that information gets agglomerated to form patterns that are recognizable and so perhaps this has something to do with the emergence of a moment where um you sort of come online because there's something like an irreducible Markov blanket that emerges by virtue of having pruned enough such that you can now make sense of the stimuli relative to some couplings in time rather than just hungry you know like there might be something um to that effect however we need to be careful again with infants what we mean about Consciousness like do you mean selfhood do you mean the capacity to reflect on self put or do you mean the capacity to um learn and act on the world which is uh closer to where we're headed because um while babies may not necessarily have this capacity to reflect on their selfhood right away they do have the capacity to react to

signals and act on the signals in order to get to a point where their preferred states are enacted so babies hungry cries food comes so there is a degree to which a baby much like a cell has a degree of Consciousness relative to our model so I'd also point you in the direction of uh Rosalind Moran's work on the Aging brain um she's got some cool work that suggests that you're well the Precision of our model increases over time uh which has to do a little bit with you know what what Mal was discussing earlier um we we know just from neuroanatomy that uh the process of pruning brain connections uh only accelerates uh through time and basically that this that means that you're disconnecting certain things from certain others right like expertise has more to do with understanding what is what is not connected to uh than understanding what is connected to what um and so uh yeah there is probably uh a kind of Developmental course whereby the model becomes sparser and sparser and sparser and sparser and you know more and more precise uh but uh you know this is a very cool direction for exploration uh I I don't know if I have anything more cogent to say at this time about it pleasure so I know Ali wanted a second girl and uh so did Samuel does anybody else have other questions because no other hands are raised at this time oh okay so we have one more hand I think I know uh actually uh you basically answered uh my other questions uh so thank you so much pleasure awesome so then uh Dallas has his hand up again Maria says her hand up and seminal has hand up so tell us first I suppose I don't know you guys like to I know it's Maria first because they just you know and then sure go ahead Maria all right so um yeah I didn't say before but I am a psychologist and master in philosophy and now I'm doing PhD in philosophy as well and it's not really a question I I would just like to to understand better what what is memory in your in your inner mother because uh I understand the the information passing and the the middle uh layers or levels that contains information but uh how exactly do we construct for for example The Narrative self or something that we remember uh from a long time or episodic memory I mean I didn't understand it very well in in the paper so if you could just shout something it's a uh it's a it's a computer science processing uh example information processing notion of memory um so uh you get some data and you need to parse it right into things that are causing that data well presumably you need to leverage some beliefs or you know knowledge about how to parse that data and that that comes in the form of information basically uh yeah uh you need to uh you know have some capacity to reduce the entropy of your observations and to kind of carve things out into things and the inner screen serves as the locus where the information is encoded so that's the cool thing about uh the quantum version of the uh markdown blanket formulation it tells you uh where all of the

information that characterizes the coupling across you know two bulks it where it lives basically I mean to get a little bit more technical the the Markov blanket itself is constituted by the degrees of freedom the couple the two systems uh together um and yeah if you uh you can think of this as an array of bits that answer yes no questions about are the are the two systems communicating across this channel uh yes or no and in some sense the state of the blanket at any given time uh it gives you all of the information that you need to characterize the coupling uh in terms of these yes no questions so all of the classical information that you need to characterize the coupling lives on the blanket and um yeah having an internal blanket means that you can have an internal store of such answers to yes no questions that you can then use to parse the sensory stream uh but in a in a kind of standard computer architecture it's just saying that you have a memory register right that you know so you yeah you need you need to store some of the information that you're processing somewhere so that you can deploy it and the idea is that it lives on this internal screen does that help and in this case it would be the hippocampus uh yeah and all well all of the inner screens in code information in this sense right the the point of having an inner screen structure at all is to kind of retain information to parse your uh your sensory stream and yeah structures like the hippocampus like these upper uh brain stem structures or what used to be called the limbic brain and so on we think that these are all at the top the idea right is that like so you have your sensory motor end then you have your prefrontal cortex which is the controller for the pre for the sensory motor and it's a controller in the sense that its main function is to inhibit activity in the sensory motor area but um if if uh if the main function of prefrontal Cortex is to go to the rest of the brain then there is also a part of the brain whose main function it is to go to the prefrontal cortex and that is these are these upper regions that we're describing um so yeah I the memory register is like live in this whole stack ultimately uh like the point about memory is less unique to this Apex structure what is unique to these top structures though is the re the right only kind of directionality that they that they don't have any context right like there there is no higher structure that can provide memory to these uh top layers so all they can do is act right and and in particular they act in a kind of covert Fashion on other layers of uh the brain uh I think would be the answer to that now did you want to add anything to that there's a so you touched on the notion of selfhood and how the narrative cell unfolds over time and leverages memory and our model basically established it with self-hood the process of selfing is because according to our model and very uh like sorry I'm sick uh very like um Thomas metzinger's model of Southwood it's a process not a thing so selfing is the process by which the internal

states are capable of mapping to the relevant external states such that they can act in such a way as to fulfill their own needs this is also very close to what Neil Seth about says about the beast machine um and so if you have nested structure in each of these nested structure has a time scale say so you have structures that are that happen on the milliseconds and sometimes scales that happen on um this the scale of a sentence and then some structures that track much longer time scales like your life goals and uh and in your ancestry and stuff like that um you find that memory is just the story The binning of the relevant information at the time scale at which it is relevant and so you can retrieve the relevant information for the shorter time scales because it acts as a prior for that for the lower time scale does that help for the narrative so yes all right thank you awesome sir go ahead about the uh going back to the band psychism problem uh in how you described consciousness I think it's very well I mean it's old in the literature that you can have information processing which is not conscious but it's mental in some sense in other animals or other or the beings which are not biological beings and but when you describe your models I know you have this level also but you also have a level of phenomenal awareness or phenomenal phenomenal Consciousness which you you are describing and I think that's the most particularly important part of the the model the phenomenal awareness part it's the model is the part of the model that can contribute to something new to me in in a sense that wasn't described so I'm glad you mentioned matzinger because Metzger is one of the few who have described Consciousness as a virtual reality and what I think is interesting about describing Consciousness as a virtual reality is that is that it makes a difference between mental information processing or unconscious information processes in those processes that are conscious because they are simulated somehow uh so an interesting thing of having a different language to speak of conscious events is that we can then explain why they are different and we can explain why we have images in our head and stuff like that and why these images seem to have causal Powers if if they have at all so when I imagine something you can make my my hand move or something like that you can describe this only with electrical information or something like that but it doesn't explain why we have different information processes with Consciousness and those without Consciousness and I know all Theory Of Consciousness have a a distinction between the unconscious and the conscious level but adding something like the metaphysics or virtual reality would explain the metaphysics of conversation for instance if you're playing a video game does the the content of that game matter fertilization or is it just the information that stored in the computer that matters I mean there's a a variable there which means a sword does it meaning a sword make

any difference at all and I think that's something that hasn't been very much discussed in philosophy so David Chalmers has just written a book recently about virtual reality but it doesn't it doesn't apply to Consciousness yet and I think if you guys find something uh in physics that's different as as you guys are trying to do is you admitted having that aim then maybe there could be some sort of a different physical explanation to conscious causation so I would like you guys to comment on that that's really interesting we um we draw on medzinger's uh work on opacity and transparency um so you know in medzinger's work just for the audience uh there are two kinds of cognitive processes there are transparent processes uh that we are not aware of as cognitive processes per se um and then there are opaque processes uh which we are so for instance uh the the metaphor comes from uh you know trans a transparent or opaque window uh and most of the time the window just lets you see through it towards uh whatever is on the other side but sometimes it's a bit grimy and then you see the window itself and so uh you know similarly there are some processes that mainly deliver things in the world and then there are other processes that deliver mental processes to us uh as as kind of the object of our perception uh in some sense uh so these um these are processes like attention uh or like paying uh attentional like processing or paying attention to your own attentional State uh these uh processes uh make other processes opaque um Terence Deacon uh the Anthropologist also talks about similar phenomena in terms of like absence absence and what he calls absential phenomena so there's a sense in which this top layer what it does is Implement uh opacification so the the top layer can only be about other layers in the stack and uh this kind of reversal of the flow of information is that we think like the the key to understanding the way that uh like mental processes are made available for introspection at all um yeah so uh that's that's how I would respond uh Mao I saw you turn your camera on uh do you have morta no no no I don't have more to answer that question cool tell us you have another question yep so um I know guys ever really discussed some of this before special on Twitter um but it can just like repeat it um because if I understood it right when applying the model to human neurobiology the article kind of sections Central neural system could you repeat the last thing you broke you broke up pretty bad sorry my laptop is really really interesting so okay um the article seems to align itself to a more modular account of the central nervous system um but in contrast to motor accounts that some you know dynamicon distributed approaches that suggests that there are no parts of the brain with the Decay functions right um but regions with various capacities that have expressed in software assembled test specific collisions so this correlation sometimes are termed like transiently

assembled local neuronal subsystems right like yeah we're big fans of like big families yeah exactly so uh this approaches the specialism is only a functional Footprints and brain functions and I've heard the result if you should fix it networks implementing specific processes but the result of high order relations between regions so my question bear in mind knowing that you guys know this kind of approaches right um do I think that the level VR construction viable model could be cash out in terms of transient correlations built on the flight such as following towns yes I mean we have a paper doing that uh from 2020 it's called neural and phenotypic representations uh under the free energy principle and it is uh one of our lesser known uh papers let me just put it in the chat quickly um and so what we show in this paper is that it's possible to simulate the form the transient formation of Markov um that basically track stimulus features uh so uh there are a lot of moving parts to what you were saying on the one hand yes uh you can totally use the Markov blanket formalism to model the transient formation and assembly of these um I would also add that there's this there's a new uh perspective on modularity that opens up via the free energy principle um the it's it's best explored in this paper by Thomas par inesh ippolito and a few others called modules or mean Fields where uh the argument uh and again this dovetails nicely with what Mike Anderson and his group are doing currently um the argument is that uh the specialization of brain regions is not functional uh it's computational and this is effectively what Mike Anderson and Company are arguing is that like each each patch of the brain is specialized in converting certain patterns into other patterns it's not like you have you know a visual region or uh you know I think Mike's example is the Broca's area which is associated with the production of language is actually activated in more non-linguistic tasks than in linguistic tasks so what that suggests is that each brain area is kind of like contributing certain stereotyped forms of computation to the overall process that's uh unfolding in the brain um and in the 300 energy principle literature you can understand that as a kind of factorization right so what and where we have two different streams in the brain the dorsal and the ventral uh which respect which are handled the what I am seeing and where I am seeing and basically the the idea is that this uh anatomical segregation of Pathways corresponds to a factorization and inference where you are and what you are are independent you could be here you could be here uh and there yeah so uh I think I've tried to yeah so that was a really good answer I'm going to try and say it a little bit differently and see uh see what sticks to the wall each hierarchical level of the brain has its like has Mark of blankets right so it's remember these things are nested um and they exchange with other levels according to the dependency structures of the mark of blankets um and so

this is like hierarchical predictive coding right so ascending prediction errors can be read as active States well descending can constitute sensory States from the point of view of a given level um so using this factorization the Maxwell just mention some brain regions become specialized to encode distinct probability factors that are statistically independent and uh this creates a sort of physiological separation of brain regions which imply that the the the information encoded in these regions is conditionally independent of that encoded in other distinct regions uh so it allows for the formation of new functions as the brain regions interact uh so we get to the point of mental action this is enabled by the existence of a subsystem that is both irreducible so there exists no further Mark of blanket partitions of the internal States and complex or expressive enough to act like a suitably flexible uh matter controller so this matter controller coordinates overt and covert action that's the last part of what Maxwell explained which can be seen as trans as forming transient connections right so it that's the point of precision uh Wayne so the next the nested scales of increasingly large ensembles of neurons basically form um coherently and so they also have their own Mark of blankets so we can generalize to different nested connectivity structures in the brain where we find uh formally identical patterns at subordinate levels and superordinate skills so does that help restructure what Maxwell said yes yes it did awesome awesome several still has his hand up I think Ali also has his hand up um Ali didn't go yet Samuel do you mind if Ali goes first sure go ahead uh thank you yeah I just wanted to make a couple of comments about the notion of self self-everancing or self-modeling that's just been discussed because uh you see sometimes in philosophical literature the sensibility of secondary properties are usually uh they're divided into the sensibility of things in itself and Sensibility of things form itself and but under these conditions some philosophers believe that real sensation would not be possible or as Arthur rimbo uh famously put it I is an other right but in some recent developments particularly in process materialism through the work of people like Thomas nail the Manuel delanda and Karen broad they've tried to somehow overcome this issue this philosophical distinction between in itself and For Itself by introducing the notion of self-receptivity uh name or in other words a capacity to be affected and to affect at the same time because uh by introducing a self-receptivity then the the sensation of each sensed point in the flow can become itself as an other to itself so basically that distinction between in itself and for itself would be dismantled effectively and that uh what that's why it can provide a more reliable way to discuss the notion of real sensation or real even subjectivity as a flow that just falls on itself and comprises both in itself and by itself and uh that's also why I believe um fep and the active

inference can provide uh even on the uh I mean more philosophical note uh more a reliable way to talk about what it means to uh to sense and be sensed at the same time I mean I I I find that very compelling I know you and Mao have been working together so I think Mao might have something more insightful than I I think that's great though I agree nice I agree I think I like linking these ideas to what we wrote in the Consciousness text by basically saying that bigger screen is the folding onto itself right it's it can be seen as a as the wrapping inside another form the Consciousness is right into the form is wrapped in the form of the mind and so you can if you partition internal and external States and the market blankets has these different kind of active and uh sensory States you can see how the folding basically happens through this Loop uh that is it isn't necessarily happening because it's not only happening at one scale does not have to be non-simultaneous like we we consider that uh that these things are um separated at times simply because uh we're thinking of one time scale um but so if we if we add this notion of interaction um we can highlight the dynamic and relational nature of Consciousness um to refer to the process where the agent and its environment actively shape one another so the Mind creating an inner screen here just reflects its understanding of the world and of itself through like basically of the world through its own filter and the filter is replica reflection of itself it's just that effectively if there are errors you need to be able to model what parts of your filter created potentially the error so yeah um you can yeah does that does that help kind of flesh out this idea of holding onto itself and incoming sensory data I think this question was for Ali or Maxwell so one of you has to decide whether I I thought it was for Ali that was great okay cool hahaha yeah that was great thanks awesome so Samuel I think you have a my final questions so yeah I was mostly talking about the heart problems courses because I I also buy the predictive processing story and I use what you guys are saying uh but one issue that I have with the productive process in the story is that we can apply it to explanations in various ways so it's sometimes like sometimes you can it seems you can tell any story with the same Theory and it's hard to know which one is true if you can tell multiple stories so I think that's a problem with um predictive processing in general and you can explain for instance for instance autism in different ways and I think people have described that your model unconsciousness how can we know that any of that is true how empirical is it and during the talk you guys mentioned something that you were gonna test something based on this which was very uh promising and I'd like you to comment on that so Maxwell has a really he has a really cool paper on the map territory fantasy fallacy which I think is part of the answer here and we've also written some uh so I'll let Maxwell dig into it probably was going

to be a very flesh out answer so I'll give a very small answer which he's going to blow out of the water afterwards but there are some things we can do to test whether the predictions made by our model actually come to fruition um so that's that's one way to to test the the empirical validity of a model but to keep in mind a model logistic model and so since you have multiple realizability effectively what really only matters is whether your model actually has any predictive power um so here let's say that we can um examine whether the brain structure and function reflects the nested and hierarchical organization which is predicted by our model or whether intervention that alter the brain structure or function lead to changes in conscious experience as we predicted um or for instance we can test which parts of the brain relate to which scale of elements that are perceived and then make predictions about the contents of Consciousness and their neural correlates to see how well it Maps so these are things that we could empirically test I'll I'll just add a bit of meta flavor to that um so remember I started with the distinction between Dynamics mechanics and principles personally I don't like the verbiage of predictive processing I don't think it's a I don't think frankly I don't think the term refers to anything specific uh when people use the term rigorously so Andy Clark introduced the term in his 2013 paper whatever next because he wanted to talk about some broad kind of approach predict prediction involving approach to explaining cognition but at the time he he didn't want to commit to any specific implementation of the idea so uh he was like here's predictive coding he was hierarchical predictive coding but as I'm sure you know predictive coding makes very specific predictions about the neural implementation of the theory so in particular predictive coding assumes that there are distinct neuronal subpopulations that encode the error and the prediction error he didn't necessarily want to commit to one version of that because who knows maybe it's the same subpopulations that you know issue predictions and prediction errors at different times and you have some kind of like time-based like rate coding or whatever uh going on um he also at the time didn't want to commit to the free energy principle per se so you know I think yeah uh if the predictive processing doesn't really denote a specific Theory and one of the problems that emerged uh is you know there's a whole bunch of literature uh presenting accounts that are consistent with predictive processing which doesn't mean anything really uh it just means hey you can kind of wave your hands around and this kind of looks like a prediction if you kind of you know look at it the right way um I think the more rigorous way to understand this is to bring it back to the math and say that there are principles uh mechanics and Dynamics so there's some uh phenomenon that we want to explain we have a formal description of it this might be like a you know a data set or

whatever else describing like the actual values of whatever system you're considering the free energy principle um is deployed to construct specific generative models uh which provide you with the equations of motion that explain that behavior right so the generative model is I mean in effect a set of linked equations of motions that you see here with these dependence relationships um and uh yeah so when you actually commit to a specific generative model then you are in a Terrain where you can bring empirical evidence to the to to kind of decide uh validate or to falsify the model so uh like I would draw your attention to you know the the free energy principle uh we know the free energy principle is true for mathematical reasons it would be as silly to try to invalidate the fep empirically as it would be to try to you know falsify calculus by trying to observe some phenomenon in the empirical world it is true mathematically and then you use that principle to derive specific models which are then testable uh so I don't I don't know if we've gone all the way here uh so this is somewhere still kind of in between an in-principle explanation and a specific model um to get to more fleshed out empirical predictions of the sort that Mao is discussing we would have to get a bit more specific which we're doing kind of in parallel I I don't feel ready to discuss the exact predictions and the ways that we'll be um discussing them this is uh or exploring them testing them this is very much work in progress I can tell you it'll involve psychedelics but uh that uh that's pretty much uh where I'm coming I'm not comfortable extending any more than that I feel like I might expose a flank that's not ready to be exposed perfect thanks for that um and that brings us to the uh and uh oh uh tell us you have your hand up again uh do you wanna hop on who has time yeah yeah I think we'll have fun for one more question yeah but Daniel you want to ask something because whatever that's two times so okay um actually I would like for guys to delve a little bit deeper and the implications of Maria's first question it's the other side of the Quan um related to the pen photo open cycle stuff the population level conscious or conscious processes in complex organisms sharing a phenotype for example like ant colonies or no more speculative sense um like um complex organisms um aligned and shared like a distributed um generative model in Niche building processes right so as Maxwell once said the other side of the um Metro selection right process so um given the sparsity and given the transient you know um conversions that these populations are in this specific moment specific process can we speculate something about conscious in this level thank you all right I think Maxwell is uh staying quiet because he expects me to sort of started answering the question correct so um we've done a lot of work in simulating large-scale systems right uh so we've simulated uh Niche uh We've simulated interaction between one and two or more actors uh

what it means for belief sharing propagation the formation of Market blankets Etc and so um I think the key the true question would be how transients are the market blankets at the level of superstructures and how does it enable those structures to be able to encode information at the level of the interaction between different scales so I do think that given our model you can assume that a system is conscious by virtue of having nested Parts which can read and write where superordinates entity or level of the the structure is right only and the lower level is also right only and through this um this sort of boundary of the entity and the interactions between the parts of the entity you have what resembles a conscious process at which you have some degree of perception um the real the really interesting question there becomes what becomes the irreducible parts so if you could be understood a little bit like a cell if you as a human your irreducibility is that your cell as a part of a larger system yeah where what's experiences at the uh at the system level uh and I think that's a really truly interesting question if we look at ants for instance ants on their own don't do much competition right but ends as a hive then become much more interesting and they behave uh in a coordinated fashion which can look as if they were making much more complex computation and you could say the same for yourselves your cells in themselves they have a very simple uh process very simple computation but together through these large-scale emerging coordinating patterns some of them become your skin some of them become your brain uh and the in the the conscious part that you are experiencing at the scale at which you can understand Consciousness currently um that's where most information most information is lost such that the irreducible part of your Consciousness is the parts that can become opaque I think I've answered most of your now I would uh maybe just add the like so uh this is a uh not fully chartered territory and I think there may be some uh agreements and disagreements I am I'm always um wary of the intentional misattribution fallacy uh where uh you know we attribute uh intentionality or mental properties to uh things that don't end up having them I think that the kind of minimal the kind of minimal intentionality pen intentionalism pen representationalism whatever you want to call it that follows from the free energy principle uh gets you uh a lot of the intelligent looking Behavior without having to pause it that there's actually something it's like to be you know a city uh or whatever what makes me skeptical of the idea that there is such a thing is that uh what what we are suggesting is that what's necessary for Consciousness in the stronger kind of agential sense is an external Markov blanket that insulates a bunch of internal States and then these internal States in turn have this kind of nested structure uh and that's what I don't think that there is in say like a city or uh like a higher order kind of cultural I'd have to

think about this so this is you're kind of pushing at the limits of where I'm comfortable thinking which is appreciated by the way thank you for the challenge questions yeah but I think there's room for disagreement and I might be wrong about this uh I don't think we just have to keep pushing to find out who's correct uh but I find what mala said uh interesting and compelling and intriguing I I am just like you know intellectually conservative at this point in my life being an old man this is this is where I think it's a little bit complicated to fully answer that question because we we have metaphorical tools but we don't have anything ready yet to go any any any further um one thing that I like to think about the only way you're ever going to experience maybe what it's like to be part of a Consciousness which is entailed by something more than you um is having connection to sort of hyper feed hive mind thing uh and being kind of pushed and pulled so if you've ever been part of like what a mob uh or if you are on Twitter and you see people um piling onto something and then you talk to them and and they're like I don't I kind of blacked out I don't know what happened because you you can't at this at this part of the hierarchy compute the entirety of what happened all you can do is understand sort of your path that was part of a larger network of web of paths which themselves um given the priors of the superstructures you're embedded in are kind of sort of going somewhere and represent the the error push of all the people that share a similar type of model that therefore constitutes a cultural Direction uh so that's that's the best way I think we can understand like hyper priors as a function of what it might be like to be a city sort of thing but that's that's as far as I I can take this metaphor for now does anyone want to give a final thought otherwise I'll just list just a few points that I found really cool about the discussion of kind of places it takes us at the end but anyone else I just want to say the the hard problem is really hard and it's called like that for a reason so uh like we were I think we're you know hopefully moving in in the direction of making this lesson serious but uh like all of this should be taken as very preliminary investigations really trying to sketch out the Bare Bones of an explanation um and yeah there is far more work ahead of us than uh behind us I think um now any final thoughts um I like to think about it in the opposite terms in terms of like the future of evolution can be understood through like what it's like to have language and what it's like to have um larger scale cognition and therefore uh yes of course this is very preliminary research but it's cool to dream awesome any guess Maria Samuel Ollie feels so any prediction on whether when the other two proofreads will come out I'm really interested in it so the uh God you know inshallah the um the mum paper will be released early July um right now reasonably um and then the other one is has to do with uh yeah extending outwards and that might be like

another year like I don't think I don't think that's that's coming soon um but hopefully revisions to the this uh paper will be coming out also over the summer um and yeah I would anticipate July hopefully we'll we'll see there's a lot going on right now so awesome all right I'll just oh say no please first no I just want to say thank you to everyone and for answering my questions and yeah thanks well thank you for your attention uh your kind comments and your interest uh it was really fun to discuss and I think you asked a lot of important questions bills eventually or anyone else uh yeah I I also wanted to uh thank you so much for having I mean uh I know that these kinds of discussions can have uh I mean lots of implications in many many other areas but it was really illuminating and clarifying so you're too kind my friend uh thank you very much um yeah this has been fun genuinely awesome all right I'll just give a few a few comments all right so first yes congratulations to all for the recent papers it's really exciting to see a kind of Continuum and integration with all these different threads um the way that we think of physics and physics of information and the history and the philosophy of the physics of information that kind of a gradient might be reimagined such that there's a clearer understanding of when it's the firmware or the kernel of in the tradition of as I see you at all writing sometimes in versus interpretations and and that's maybe situational but that's a gray area um reminds me of the instrumentalism realism debates of 2021 your with the map of the territory which has been developed out so much further and just the ability especially in a situation in a model specific way to separate out the intelligence or information structure or the behavior from any kind of qualia type discussion um the recent developments with uh Chris fields at all and the waste energy and where free energy goes and the kind of saying like one has forgotten more than someone else has learned so just the idea that to have awareness there has to be that discardle and then that might be related to the irreversibility and the time Consciousness that also you've been uh developing the eco-evo Devo components and just multi-scale biology connecting it to traditions of thought and work in biology which is big but that's kind of how we retro grandparent back into biology promisingly with the ways that multi-sale systems are already considered and then also just the recent developments in category Theory and the Primacy of the relational and the physics as information processing and then the way that that enables very knowledgeable individuals like you all to come from a lot of the philosophy and the history and all these other threads to come together with the actual modeling in a very satisfying way so thank you all for the stream and really looking forward to the next steps bye thank you bye-bye thanks